Sahaj Bhadja

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Summary

Software Engineer with 2+ years shipping **backend systems**, **SQL/PostgreSQL**-heavy debugging, and production reliability fixes, skilled in **Python** and **AI-assisted engineering workflows**. Seeking a support-engineering role bridging customer escalation triage with hands-on backend development.

Education

University of Southern California, Los Angeles, CA **August 2024 – May 2026**
Master of Science in Computer Science *GPA: 3.75/4*
Coursework: Database Systems, Analysis of Algorithms, Web Technologies, Software Engineering

Pandit Deendayal Energy University, Gandhinagar, India **November 2020 – May 2024**
Bachelor of Technology in Computer Science and Engineering *GPA: 3.88/4*
Coursework: Operating Systems, Software Engineering, Data Retrieval, Object-Oriented Programming

Experience

USC Dornsife **Los Angeles, CA**
Software Engineer *September 2025 – Present*

- **Triaged** and **resolved production incidents** across a multi-tenant **PostgreSQL** and **SQL Server** platform serving 200+ users, writing targeted **SQL** fixes and validation checks that reduced query latency by 35%.
- **Debugged** a fault-tolerant async LLM orchestration service built with FastAPI background workers, isolating job-state failures across 1K+ AI jobs and shipping fixes that tripled throughput under concurrent load.

Grad8 **Los Angeles, CA**
Software Engineer Intern *June 2025 – August 2025*

- Used **AI-assisted tooling** (GPT-4o, Claude, Qwen-VL) to **triage** grading-accuracy and OCR failures in a Flask/React platform, shipping fixes that cut feedback turnaround time by 60% for 150+ instructors.
- **Investigated** and **fixed** concurrency bugs in a multithreaded batch-grading pipeline handling 10K+ daily requests, adding fault-tolerant error handling and cost-tracking logs that stabilized API performance under peak load.

Aasma Technology Solutions **India**
Software Developer Engineer - (SecureSync) *August 2023 – July 2024*

- Owned **technical escalations** for a cross-platform microservice data-sharing product, reproducing defects via Jenkins-driven nightly test automation and shipping **PostgreSQL**/API fixes that improved response times by 30% for low-bandwidth users.
- **Root-caused** recurring ETL and API failures using **AWS** Glue and S3, writing PyTest/Mockito coverage and redesigned pipelines that boosted efficiency 68% while cutting technical debt 20% across Agile sprints.

Aasma Technology Solutions **India**
Full Stack Developer Intern - (VibrantChem) *May 2023 – August 2023*

- **Debugged** real-time synchronization issues in a MERN-based CRM using WebSockets and **Azure SQL**, resolving conflicts across 2M+ synced records and reducing sync errors while collaborating in Agile sprint reviews.
- **Monitored production health** with Sentry and Azure Monitor, **root-causing** API and database performance issues that, once fixed, reduced resource wastage by 42% and improved lead conversion by 15%.

LST Capitals **India**
Web Developer Intern *November 2022 – January 2023*

- **Communicated** directly with the client to gather requirements and present the system architecture, then built Flask APIs for booking and payments delivering sub-200ms latency across 50K+ booking records.
- **Diagnosed** webhook and payment-state inconsistencies across a Firebase-backed booking system, shipping idempotent fixes that cut manual refund handling by roughly 70% for the client's operations team.

Skills

Languages: Go (Golang), Python, Java, JavaScript/TypeScript, SQL, Bash/Shell, C++
Backend & APIs: REST APIs, FastAPI, Flask, Node.js, Express.js
Databases: PostgreSQL, MySQL, SQL Server, Azure SQL, MongoDB, Firebase
Cloud & Infrastructure: AWS (EC2, S3, RDS, Glue), Azure, Docker, Kubernetes, CI/CD, Jenkins
Engineering Practices: Unix Shell Scripting, Root-Cause Analysis, Secrets Management, LLM-Assisted Debugging, Agile/Scrum, Query Optimization, Code Review

Projects

Chat2SQL Code | *Java, TypeScript, SQL, Docker*

- Built an **AI-powered SQL** workspace on the Chat2DB codebase across Java and TypeScript, **debugging** schema-sync and query-generation issues across 10+ databases including **PostgreSQL**, MySQL, and SQL Server.

Big-Brain Live | *PostgreSQL, LLaMA-2, FAISS, GCP*

- Built and **debugged** a semantic-search SaaS platform over 50K+ documents using LLaMA-2, FAISS, and **PostgreSQL** on GCP, fixing retrieval-latency regressions and shipping fixes that cut document-analysis time by 60%.

HMIS Live | *MERN, ABDM APIs, Sentry, TypeScript*

- Built a healthcare platform integrating India's ABDM APIs with consent-based access controls, using Sentry monitoring to catch and fix **production** errors while **securely** handling sensitive patient data for 100+ daily appointments.